

Year 8 Science – Booklet 10: Biology

Respiration & Circulation — ANSWERS

Note: every page in this booklet covers the lungs, breathing and respiration only – there is no circulation/heart content in the pages provided, so none is answered below.

Quick Test (p.19)

1. At the alveoli, in the lungs.
2. There are millions of them, giving a large surface area; they are surrounded by lots of blood capillaries; their walls are thin and their lining is moist, so gases can dissolve and diffuse easily across the thin membrane.
3. Respiration is a chemical reaction that breaks down glucose from food to release energy, using oxygen (aerobic respiration).
4. Inside the cytoplasm of (every living) cell.
5. Carbon dioxide and water.

Section A – Multiple Choice (5 marks)

1. b) a chemical reaction that uses glucose and oxygen
2. c) carbon dioxide, water and energy
3. b) thin walled alveoli
4. a) bronchi
5. d) reduce in surface area

Section B (9 marks)

1. Air sac diagram

- a). Alveoli (alveolus, singular).
- b). Bronchiole.
- c). Gas exchange.
- d). Oxygen (diffuses from the air sac into the blood) and carbon dioxide (diffuses from the blood into the air sac).
- e). They have a large surface area (there are millions of alveoli) and are surrounded by a dense network of blood capillaries (also accept: a moist lining, to help gases dissolve).

2. Breathing in – cross out the incorrect words (4 marks)

When you breathe in the intercostal muscles and the diaphragm muscle **contract**. The ribs move **up** and **out**. The volume in the thorax **increases** and air rushes into the lungs.

Section C (7 marks)

1. Respiratory system diagram

- a). Trachea → bronchus → bronchioles → alveoli.
- b). At the alveoli.
- c). Oxygen.
- d). Carbon dioxide.

2. *Respiration*

- a). Glucose (from food, via digestion).
- b). Inside the cytoplasm of cells.
- c). Any of: making muscles work, growth and repair of cells, chemical reactions in cells, maintaining body temperature (in warm-blooded animals).